

BMAT205L – Discrete Mathematics and Graph Theory

MODULE 1: MATHEMATICAL LOGIC

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Discrete Mathematics

- Discrete Mathematics deals with the study of Mathematical structures.
- It deals with objects that can have distinct separate values. It is also called Decision Mathematics or finite Mathematics.
- It is the study of mathematical structures that are fundamentally discrete in nature and it does not require the notion of continuity.
- Objects that are studied in discrete mathematics are largely countable sets such as formal languages, integers, finite graphs, and so on.
- Due to its application in Computer Science, it has become popular in recent decades. It is used in programming languages, software development, cryptography, algorithms etc.
- Discrete Mathematics covers some important concepts such as set theory, graph theory, logic, permutation and combination as well.

Mathematical Statements

Investigate!

While walking through a fictional forest, you encounter three trolls guarding a bridge. Each is either a knight, who always tells the truth, or a knave, who always lies. The trolls will not let you pass until you correctly identify each as either a knight or a knave. Each troll makes a single statement:

Troll 1: If I am a knave, then there are exactly two knights here.

Troll 2: Troll 1 is lying.

Troll 3: Either we are all knaves or at least one of us is a knight.

Which troll is which?

Propositions

- A declarative sentence which is true or false, but not both, is called a proposition (or statement)
- Sentences which are exclamatory, interrogative or imperative in nature are not propositions
- Lower case letters such as $p, q, r, \dots$ are used to denote propositions
- For example,

1. New Delhi is the capital city of India
2. How beautiful is Rose?
3. $2+2=3$
4. What time is it?
5. $x + y = z$
6. Take a cup of coffee

(2), (4) and (6) are obviously not propositions as they are not declarative in nature

(1) and (3) are propositions, but (5) is not

Truth value of proposition

- If proposition is true, we say that the truth value of that proposition is true, denoted by T or 1.
- If the proposition is false, we say that the truth value of the proposition is false, denoted by F or 0.

Atomic or Molecular propositions

- Propositions which do not contain any of the logical operators or connectives are called atomic (primary or primitive) propositions
- Many mathematical statements which can be constructed by combining one or more atomic statements using connectives are called molecular or compound propositions
- The truth value of a compound proposition depends on those of sub-propositions and the way in which they are combined using connectives
- The area of logic that deals with propositions is called propositional logic or propositional calculus

Connectives

- When p and q are any two propositions, the proposition " p and q " denoted by $p \wedge q$ and called the **conjunction** of p and q is defined as the compound proposition that is true when both p and q are true and is false otherwise.
- The proposition " p or q " denoted by $p \vee q$ and called the **disjunction** of p and q is defined as the compound proposition that is false when both p and q are false and is true otherwise.
- Given any proposition, p , another proposition formed by writing "It is not the case that" or "It is false that" before p or by inserting the word 'not' suitably in p is called the **negation** of p and denoted by $\neg p$ (read as 'not p '). $\neg p$ is also denoted as p' , $\bar{p}$ and $\sim p$. If p is true, then $\sim p$ is false and vice versa.
- Example for negation: p : New Delhi is in India, then $\sim p$ is one of the following statements
 - it is not the case that New Delhi is in India
 - It is false that New Delhi is in India
 - New Delhi is not in India

Order of Precedence for Logical Connectives

We will generally use parentheses to specify the order in which logical operators in a compound propositions are to be applied

For example, $(p \vee q) \wedge (\sim r)$ is the conjunction of $p \vee q$ and $\sim r$. However to avoid the use of an excessive number of parentheses, we adopt an order of precedence for the logical operators, given as follows:

1. The negation operator has precedence over all other logical operators. Thus $\sim p \wedge q$ means $(\sim p) \wedge q$ not $\sim(p \wedge q)$
2. The conjunction operator has precedence over the disjunction operator. Thus $p \wedge q \vee r$ means $(p \wedge q) \vee r$, but not $p \wedge (q \vee r)$
3. The conditional and biconditional operators $\rightarrow$ and $\leftrightarrow$ have lower precedence than other operators. Among them $\rightarrow$ has precedence over $\leftrightarrow$

Conditional Propositions

- If p and q are propositions, the compound proposition “*if p , then q* ”, that is denoted by $p \rightarrow q$ is called a **conditional proposition**, which is **false** when **p is true and q is false** and true otherwise.
- If this condition proposition, p is called the ***hypothesis or premise*** and q is called the ***conclusion or consequence***.
- The alternative terminologies use to express $p \rightarrow q$ are the following:
 1. p implies q
 2. p only if q
 3. q if p or q when p
 4. q follows from p
 5. p is sufficient for q or a sufficient condition for q is p
 6. q is necessary for p or a necessary condition for p is q

Biconditional proposition

- If p and q are propositions, the compound proposition “ **p if and only if q** ”, that is denoted by $p \leftrightarrow q$ is called a **biconditional proposition**, which is true when p and q have the same truth values and is false otherwise.
- It is easily verified that $p \leftrightarrow q$ is true when both the conditionals $p \rightarrow q$ and $q \rightarrow p$ are true. This is the reason for the symbol $\leftrightarrow$ which is combination of $\rightarrow$ and $\leftarrow$.
- Alternatively, $p \leftrightarrow q$ is also expressed as ‘ p iff q ’ and p is necessary and sufficient for q

Tautology and Contradiction

- A compound proposition $P = P(p_1, p_2, \dots, p_n)$ where $p_1, p_2, \dots, p_n$ are variables (elemental propositions), is called a **tautology**, if it is true for every truth assignment for $p_1, p_2, \dots, p_n$.
- P is **contradiction**, if it is false for every truth assignment for $p_1, p_2, \dots, p_n$.
- For example, $p \vee \sim p$ is a tautology, whereas $p \wedge \sim p$ is a contradiction
- Note:
- The negation of a tautology is a contradiction and the negation of a contradiction is a tautology.
- If $P = P(p_1, p_2, \dots, p_n)$ is a tautology, then $P = P(q_1, q_2, \dots, q_n)$ is also a tautology, where $q_1, q_2, \dots, q_n$ are any set of propositions. This is known as the principle of substitution. For example $p \vee \sim p$ is a tautology, $((p \vee q) \wedge r) \vee \sim((p \vee q) \wedge r)$ is also a tautology.
- If a proposition is neither a tautology nor a contradiction it is called a **contingency**

Problems

Construct a truth table for each of the following compound propositions:

1. $(p \vee q) \rightarrow (p \wedge q)$
2. $(p \rightarrow q) \rightarrow (q \rightarrow p)$
3. $(q \rightarrow \sim p) \leftrightarrow (p \leftrightarrow q)$
4. $(p \leftrightarrow q) \leftrightarrow ((p \wedge q) \vee (\sim p \wedge \sim q))$
5. $(\sim p \leftrightarrow \sim q) \leftrightarrow (p \leftrightarrow q)$

Problems

Construct a truth table for each of the following compound propositions. Determine which of the following propositions are tautologies, contradictions and Contingency using truth table.

1. $(p \rightarrow (q \rightarrow r)) \rightarrow ((p \rightarrow q) \rightarrow (p \rightarrow r))$
2. $\sim(p \vee (q \wedge r)) \leftrightarrow ((p \vee q) \wedge (p \rightarrow r))$
3. $(\sim p \leftrightarrow \sim q) \leftrightarrow (q \leftrightarrow r)$
4. $(p \rightarrow (q \rightarrow s)) \wedge (\sim r \vee p) \wedge q$
5. $((p \rightarrow q) \rightarrow r) \rightarrow s$
6. $(\sim p \wedge (p \rightarrow q)) \rightarrow \sim p$
7. $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$
8. $\sim(q \rightarrow r) \wedge r \wedge (p \rightarrow q)$
9. $((p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)) \rightarrow r$

Equivalence of Propositions

- Two compound propositions $A(p_1, p_2, \dots, p_n)$ and $B(p_1, p_2, \dots, p_n)$ are said to be logically equivalent or simply equivalent, if they have identical truth tables.
- If truth value of A is equal to the truth value of B for every one of the 2^n **possible sets** of truth values assigned to $p_1, p_2, \dots, p_n$
- The equivalence of two propositions A and B is denoted as $A \Leftrightarrow B$ or $A \equiv B$ which is read as '**A is equivalent to B**' ($\Leftrightarrow$ or $\equiv$ is not connective).
- For example, let us consider the truth tables of $\sim(p \vee q)$ and $\sim p \wedge \sim q$. The final columns in the truth tables are identical. Hence, $\sim(p \vee q) \equiv \sim p \wedge \sim q$.
- Note: $A \Leftrightarrow B$ is tautology. When A and B are equivalent. Conversely, A and B are equivalent, when $A \Leftrightarrow B$ is tautology.
- $p \rightarrow q \equiv (\sim p \vee q)$, since $p \rightarrow q \Leftrightarrow (\sim p \vee q)$ is a tautology.

Duality Law

- The dual of a compound proposition that contains only the logical operators $\vee$, $\wedge$ and $\sim$ is the proposition obtained by replacing each $\vee$ by $\wedge$, each $\wedge$ by $\vee$, each T by F and each F by T are special variables representing compound propositions that are tautologies and contradictions respectively.
- The dual of a proposition A is denoted by A^* .
- Duality Theorem: If $A(p_1, p_2, \dots, p_n) \equiv B(p_1, p_2, \dots, p_n)$, where A and B are compound propositions, then $A^*(p_1, p_2, \dots, p_n) \equiv B^*(p_1, p_2, \dots, p_n)$.
- De Morgan's Law: $\sim(p \vee q) = \sim p \wedge \sim q$, $p \vee q = \sim(\sim p \wedge \sim q)$, $p \wedge q = \sim(\sim p \vee \sim q)$
- $\sim(A(p_1, p_2, \dots, p_n)) = A^*(p_1, p_2, \dots, p_n)$

Algebra of Propositions

- A proposition in a compound proposition can be replaced by one that is equivalent to it without changing the truth value of the compound proposition.
- By this, new equivalents (or laws) can be constructed
- For example, we have proved that $p \rightarrow q \equiv \sim p \vee q$.
- Using this equivalence, we get another equivalence $p \rightarrow (q \rightarrow r) \equiv (p) \rightarrow (\sim q \vee r)$

Sl. No.	Name of the law	Primal form	Dual form
1.	Idempotent Law	$p \vee p \equiv p$	$p \wedge p \equiv p$
2.	Identity Law	$p \vee F \equiv p$	$p \wedge T \equiv p$
3.	Dominant Law	$p \vee T \equiv T$	$p \wedge F \equiv F$
4.	Complement Law	$p \vee \sim p \equiv T$	$p \wedge \sim p \equiv F$
5.	Commutative Law	$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$
6.	Associative Law	$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
7.	Distribution Law	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
8.	Absorption Law	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
9.	De Morgan's Law	$\sim(p \vee q) \equiv \sim p \wedge \sim q$	$\sim(p \wedge q) \equiv \sim p \vee \sim q$

Equivalences involving conditionals

1. $p \rightarrow q \equiv \sim p \vee q$
2. $p \rightarrow q \equiv \sim q \rightarrow \sim p$
3. $p \vee q \equiv \sim p \rightarrow q$
4. $p \wedge q \equiv \sim(p \rightarrow \sim q)$
5. $\sim(p \rightarrow q) \equiv p \wedge \sim q$
6. $(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$
7. $(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$
8. $(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$
9. $(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$

Equivalences involving biconditionals

1. $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
2. $p \leftrightarrow q \equiv \sim p \leftrightarrow \sim q$
3. $p \leftrightarrow q \equiv (p \wedge q) \vee (\sim p \wedge \sim q)$
4. $\sim(p \leftrightarrow q) \equiv p \leftrightarrow \sim q$

Tautological Implication

- A compound proposition $A(p_1, p_2, \dots, p_n)$ is said to **tautologically imply** or simply imply the compound proposition $B(p_1, p_2, \dots, p_n)$, if **B is true whenever A is true** or equivalently if and only if **$A \rightarrow B$ is tautology**. This is denoted by $A \Rightarrow B$, read as "*A implies B*"
- $\Rightarrow$ is not a connective and $A \Rightarrow B$ is not a proposition
- For example, $p \Rightarrow p \vee q$, $p \vee q$ is true whenever p is true and that $p \rightarrow (p \vee q)$ is a tautology.

•Implications:

1. $p \wedge q \Rightarrow p$

2. $p \wedge q \Rightarrow q$

3. $p \Rightarrow p \vee q$

4. $\sim p \Rightarrow p \rightarrow q$

5. $q \Rightarrow p \rightarrow q$

6. $\sim(p \rightarrow q) \Rightarrow p$

7. $\sim(p \rightarrow q) \Rightarrow \sim q$

8. $p \wedge (p \rightarrow q) \Rightarrow q$

9. $\sim q \wedge (p \rightarrow q) \Rightarrow \sim p$

10. $\sim p \wedge (p \vee q) \Rightarrow q$

11. $(p \rightarrow q) \wedge (q \rightarrow r) \Rightarrow p \rightarrow r$

12. $(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r) \Rightarrow r$

Problem 1: Without using truth table, prove the following: $(\sim p \vee q) \wedge (p \wedge (p \wedge q)) \equiv p \wedge q$

Proof:

$(\sim p \vee q) \wedge (p \wedge (p \wedge q)) \equiv (\sim p \vee q) \wedge ((p \wedge p) \wedge q)$, by associative law

$\equiv (\sim p \vee q) \wedge (p \wedge q)$ by idempotent law

$\equiv (p \wedge q) \wedge (\sim p \vee q)$, by commutative law

$\equiv ((p \wedge q) \wedge \sim p) \vee ((p \wedge q) \wedge q)$, by distributive law

$\equiv (\sim p \wedge (p \wedge q)) \vee ((p \wedge q) \wedge q)$, by commutative law

$\equiv ((\sim p \wedge p) \wedge q) \vee (p \wedge (q \wedge q))$, by associative law

$\equiv (F \wedge q) \vee (p \wedge q)$, by complement and idempotent law

$\equiv F \vee (p \wedge q)$, by dominant law

$\equiv p \wedge q$, by dominant law

Problem 2: Without using truth table, prove the following: $p \rightarrow (q \rightarrow p) \equiv \sim p \rightarrow (p \rightarrow q)$

Proof:

$$\text{L.H.S } p \rightarrow (q \rightarrow p) \equiv \sim p \vee (q \rightarrow p)$$

$$\equiv \sim p \vee (\sim q \vee p)$$

$$\equiv \sim p \vee (p \vee \sim q)$$

$$\equiv (\sim p \vee p) \vee q$$

$$\equiv T \vee q$$

$$\equiv T$$

$$\text{R.H.S } \sim p \rightarrow (p \rightarrow q) \equiv p \vee (p \rightarrow q)$$

$$\equiv p \vee (\sim p \vee q)$$

$$\equiv (\sim p \vee p) \vee q$$

$$\equiv T \vee q$$

$$\equiv T$$

$$\text{L.H.S}=\text{R.H.S}$$

Hence Proved

Problem 3: Without using truth table, prove the following: $\sim(p \leftrightarrow q) \equiv (p \vee q) \wedge \sim(p \vee q) \equiv (p \wedge \sim q) \vee (\sim p \wedge q)$

Proof:

$$\text{L.H.S : } \sim(p \leftrightarrow q) \equiv \sim((p \rightarrow q) \wedge (q \rightarrow p))$$

$$\equiv \sim((\sim p \vee q) \wedge (\sim q \vee p))$$

$$\equiv \sim(((\sim p \vee q) \wedge \sim q) \vee ((\sim p \vee q) \wedge p)), \text{ by distribution law}$$

$$\equiv \sim(((\sim p \wedge \sim q) \vee (q \wedge \sim q)) \vee ((\sim p \wedge p) \vee (q \wedge p))), \text{ by distribution law}$$

$$\equiv \sim(((\sim p \wedge \sim q) \vee F) \vee (F \vee (q \wedge p)))$$

$$\equiv \sim((\sim p \wedge \sim q) \vee (q \wedge p)), \text{ by identity law}$$

$$\equiv \sim(\sim(p \vee q) \vee (q \wedge p)), \text{ by De Morgan's law}$$

$$\equiv (p \vee q) \wedge \sim(q \wedge p), \text{ by De Morgan's law}$$

$$\equiv (p \vee q) \wedge (\sim q \vee \sim p), \text{ by De Morgan's law}$$

$$\equiv ((p \vee q) \wedge \sim q) \vee ((p \vee q) \wedge \sim p), \text{ by distribution law}$$

$$\equiv ((p \wedge \sim q) \vee (q \wedge \sim q)) \vee ((p \wedge \sim p) \vee (q \wedge \sim p)), \text{ by distribution law}$$

$$\equiv ((p \wedge \sim q) \vee F) \vee (F \vee (q \wedge \sim p))$$

$$\equiv (p \wedge \sim q) \vee (q \wedge \sim p), \text{ by identity law}$$

$$\equiv (p \wedge \sim q) \vee (\sim p \wedge q), \text{ by commutative law}$$

Problem 4: Without constructing the truth tables, prove the following:

(i) $\sim p \rightarrow (q \rightarrow r) \equiv q \rightarrow (p \vee r)$

(ii) $p \rightarrow (q \rightarrow r) \equiv p \rightarrow (\sim q \vee r) \equiv (p \wedge q) \rightarrow r$

(iii) $((p \vee q) \wedge \sim(\sim p \wedge (\sim q \vee \sim r))) \vee (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r)$ is a tautology

Solution:

(i) $\sim p \rightarrow (q \rightarrow r) \equiv p \vee (q \rightarrow r)$

$\equiv p \vee (\sim q \vee r)$

$\equiv (p \vee \sim q) \vee r$, by associative law

$\equiv (\sim q \vee p) \vee r$, by commutative law

$\equiv (\sim q) \vee (p \vee r)$, by associative law

$\equiv q \rightarrow (p \vee r)$

$$(ii) p \rightarrow (q \rightarrow r) \equiv p \rightarrow (\sim q \vee r) \equiv (p \wedge q) \rightarrow r$$

Solution

$$p \rightarrow (q \rightarrow r) \equiv \sim p \vee (q \rightarrow r)$$

$$\equiv \sim p \vee (\sim q \vee r)$$

$$\equiv (\sim p \vee \sim q) \vee r, \text{ by associative law}$$

$$\equiv \sim(p \wedge q) \vee r, \text{ by De Morgan's law}$$

$$\equiv (p \wedge q) \rightarrow r$$

(iii) $((p \vee q) \wedge \sim(\sim p \wedge (\sim q \vee \sim r))) \vee (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r)$ is a tautology

$$((p \vee q) \wedge \sim(\sim p \wedge (\sim q \vee \sim r))) \vee (\sim p \wedge \sim q) \vee (\sim p \wedge \sim r)$$

$$\equiv \left((p \vee q) \wedge \sim(\sim(p \vee (q \wedge r))) \right) \vee \sim(p \vee q) \vee \sim(p \wedge r), \text{ by De Morgan's law}$$

$$\equiv \left((p \vee q) \wedge (p \vee (q \wedge r)) \right) \vee \sim(p \vee q) \vee \sim(p \wedge r)$$

$$\equiv ((p \vee q) \wedge (p \vee q) \wedge (p \vee r)) \vee (\sim(p \vee q) \vee \sim(p \wedge r)), \text{ by distributive law}$$

$$\equiv ((p \vee q) \wedge (p \vee r)) \vee (\sim((p \vee q) \wedge (p \wedge r))), \text{ by idempotent law and De Morgan's law}$$

$$\equiv T$$

Problem 5: Prove the following equivalences by proving the equivalences of the duals:

$$(i) \sim((\sim p \wedge q) \vee (\sim p \wedge \sim q)) \vee (p \wedge q) \equiv p$$

$$(ii) (p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$$

$$(iii) (p \wedge (p \leftrightarrow q)) \rightarrow q \equiv T$$

Solution:

$$(i) \sim((\sim p \wedge q) \vee (\sim p \wedge \sim q)) \vee (p \wedge q) \equiv p$$

The dual of the given equivalence is

$$\sim((\sim p \vee q) \wedge (\sim p \vee \sim q)) \wedge (p \vee q) \equiv p$$

$$\text{Proof: L.H.S: } \sim((\sim p \vee q) \wedge (\sim p \vee \sim q)) \wedge (p \vee q)$$

$$\equiv \sim\left((\sim p \vee (q \wedge \sim q))\right) \wedge (p \vee q), \text{ by distribution law}$$

$$\equiv \sim\left((\sim p \vee (F))\right) \wedge (p \vee q)$$

$$\equiv \sim(\sim p) \wedge (p \vee q), \text{ by identity law}$$

$$\equiv p \wedge (p \vee q)$$

$$\equiv p, \text{ by absorption law}$$

$$(ii) (p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$$

Solution

$$(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$$

$$\Rightarrow \sim(p \vee q) \vee r \equiv (\sim p \vee r) \wedge (\sim q \vee r)$$

The dual of equivalence is $\sim(p \wedge q) \wedge r \equiv (\sim p \wedge r) \vee (\sim q \wedge r)$

Proof:

$$\text{L.H.S} \equiv \sim(p \wedge q) \wedge r$$

$$\equiv (\sim p \vee \sim q) \wedge r, \text{ by De Morgan's Law}$$

$$\equiv (\sim p \wedge r) \vee (\sim q \wedge r), \text{ by distribution law}$$

Hence Proved

$$(iii) (p \wedge (p \leftrightarrow q)) \rightarrow q \equiv T$$

Solution:

$$(p \wedge (p \leftrightarrow q)) \rightarrow q \equiv T$$

$$\Rightarrow (p \wedge ((p \rightarrow q) \wedge (q \rightarrow p))) \rightarrow q \equiv T$$

$$\Rightarrow (p \wedge ((\sim p \vee q) \wedge (\sim q \vee p))) \rightarrow q \equiv T$$

$$\Rightarrow \sim (p \wedge ((\sim p \vee q) \wedge (\sim q \vee p))) \vee q \equiv T$$

$$\text{Dual of this } \sim (p \vee ((\sim p \wedge q) \vee (\sim q \wedge p))) \wedge q \equiv F$$

$$\begin{aligned}
\text{L.H.S} &\equiv \sim \left((p \vee ((\sim p \wedge q)) \vee (\sim q \wedge p)) \right) \wedge q, \text{ by associative law} \\
&\equiv \sim (((p \vee \sim p) \wedge (p \vee q)) \vee (\sim q \wedge p)) \wedge q, \text{ by distributive law} \\
&\equiv \sim ((T \wedge (p \vee q)) \vee (\sim q \wedge p)) \wedge q \\
&\equiv \sim ((p \vee q) \vee (\sim q \wedge p)) \wedge q, \text{ by identity law} \\
&\equiv \sim (((p \vee q) \vee \sim q) \wedge ((p \vee q) \vee p)) \wedge q, \text{ by distributive law} \\
&\equiv \sim ((p \vee T) \wedge (p \vee q)) \wedge q, \text{ by idempotent and complemen law} \\
&\equiv \sim (T \wedge (p \vee q)) \wedge q, \text{ identity law} \\
&\equiv \sim (p \vee q) \wedge q \\
&\equiv (\sim p \wedge \sim q) \wedge q \\
&\equiv \sim p \wedge F \\
&\equiv F
\end{aligned}$$

Problem 6: Prove the following implications by using the truth tables:

(i) $p \rightarrow (p \rightarrow r) \Rightarrow (p \rightarrow q) \rightarrow (p \rightarrow r)$

(ii) $(p \rightarrow (q \rightarrow s)) \wedge (\sim r \vee p) \wedge q \Rightarrow r \rightarrow s$

W.K.T $A \Rightarrow B$, if and only if $A \rightarrow B$ tautology.

Problem 7: Prove the following implications without using truth tables:

(i) $(p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r) \Rightarrow r$

(ii) $((p \vee \sim p) \rightarrow q) \rightarrow ((p \vee \sim p) \rightarrow r) \Rightarrow q \rightarrow r$

Normal Forms

- To determine whether a given compound proposition $A(p_1, p_2, \dots, p_n)$ is a tautology or a contradictor or at least satisfiable and whether two given compound propositions $A(p_1, p_2, \dots, p_n)$ and $B(p_1, p_2, \dots, p_n)$ are equivalent, we have to construct truth tables and compare them
- $A(p_1, p_2, \dots, p_n)$ is said to be **satisfiable**, if it has the **truth value T** for **at least one combination** of the truth value of $p_1, p_2, \dots, p_n$ of primary propositions.
- But the construction of truth table may not be practical, when the number of primary propositions $p_1, p_2, \dots, p_n$ increases. **A better method is to reduce A and B to some standard forms, called normal forms** and use them for deciding the nature of A or B and for comparing A and B .
- There are two types of normal forms – disjunctive normal form and conjunctive normal form
- We shall use the ‘word’ product in place of ‘conjunction’ and ‘sum’ in place ‘disjunction’ hereafter

Disjunctive and Conjunctive Normal

- A product of the variables and their negations is called an elementary product
- Similarly, a sum of the variables and their negations is called an elementary sum.
- For example, p , $\sim p$, $p \wedge \sim p$, $\sim p \wedge q$, $p \wedge \sim q$ and $\sim p \wedge \sim q$ are some elementary products in 2 variables
- q , $\sim q$, $p \vee q$, $p \vee \sim q$, $\sim p \vee \sim q$ are some elementary products in 2 variables.
- A compound proposition which consists of a **sum of elementary product** and which is equivalent to a given proposition is called a **disjunctive normal form(DNF)** of given proposition
- A formula which consists of a **product of elementary sums** and which is equivalent to a given formula is called **conjunctive normal form (CNF)** of the given formula.

Procedure to obtain the DNF or CNF of a given formula

Step 1 : If the connectives $\rightarrow$ and $\leftrightarrow$ are present in the given formula they are replaced by $\wedge, \vee$ and $\sim$.

Eg. $p \rightarrow q$ is replaced by $\sim p \vee q$ and $p \leftrightarrow q$ is replaced by either $(p \wedge q) \vee (\sim p \wedge \sim q)$ or $(\sim p \vee q) \wedge (\sim q \vee p)$

Handwritten notes in red:
 $p \rightarrow q \wedge (q \rightarrow p)$
 $\sim(p \wedge q)$

Step 2: If the negation is present before the given formula or a part of the given formula. De Morgan's laws are applied so that the negation is brought before the variables only.

Step 3: If necessary, the distribution law and the idempotent law are applied.

Step 4: If there is an elementary product which is equivalent to the truth value F in the DNF, it is omitted. Similarly, if there is an elementary sum which is equivalent to the truth value T in the CNF, it is omitted.

Example

- DNF of $q \rightarrow (q \rightarrow p)$ is given by

$$q \rightarrow (q \rightarrow p) \equiv \sim q \vee (q \rightarrow p)$$

$$\equiv \sim q \vee (\sim q \vee p)$$

$$\equiv (\sim q \vee \sim q) \vee p \text{ (by association law)}$$

$$\equiv \sim q \vee p \text{ (by idempotent law)}$$

- CNF of $\sim(p \vee q) \leftrightarrow (p \wedge q)$ is given by

$$\sim(p \vee q) \leftrightarrow (p \wedge q) \equiv (\sim(p \vee q) \wedge (p \wedge q)) \wedge (\sim(\sim(p \vee q)) \wedge (\sim(p \wedge q)))$$

$$\equiv (\sim p \wedge \sim q) \wedge (p \wedge q) \vee (p \vee q) \wedge (\sim p \vee \sim q)$$

$$\equiv (p \wedge \sim p) \wedge (q \wedge \sim q) \vee (p \vee q) \wedge (\sim p \vee \sim q)$$

$$\equiv F \wedge F \vee (p \vee q) \wedge (\sim p \vee \sim q)$$

$$\equiv (p \vee q) \wedge (\sim p \vee \sim q)$$

p	$\sim p$	$\wedge$
T	F	F
F	T	F

$\sim(\sim a) \equiv a$

Principal disjunctive and principal conjunctive normal forms

- Given a number of variables, **the product (or conjunctions) in which each variable or its negation, but not both occurs only once** are called the **minterms**. For two variable p and q , the possible minterms are $p \wedge q, p \wedge \sim q, \sim p \wedge q$ and $\sim p \wedge \sim q$
- We note that there are 2^n minterms for n variables.
- Given a number of variables, **the sum in which each variable or its negation but not both occurs only once** are called **maxterms**. Maxterms are simply the duals of minterms
- The formula consisting of **disjunctions of minterms** in the variables only and equivalent to a given formula is known as its **principal disjunctive normal form (PDNF)** or its sum of products canonical form of the given formula.
- Similarly, a formula consisting of **conjunction of maxterms** in the variables only and equivalent to given formula is known as its **principal conjunctive normal form (PCNF)** or its product of sums canonical form.

-
- In order to obtain a DNF of the formula by using the procedure given above. To get the minterms in the disjunctions, the missing factors are introduced through the complement law ($p \vee \sim p = T$) and then applying the distributive law.
 - Identical minterms appearing in the disjunctions are then deleted, as $p \vee p = p$.
 - A similar procedure with necessary modifications is adopted to get PCNF of a formula.
 - In order to verify whether two given formulas are equivalent, we may obtain either PDNF or PCNF of both the formulas and compare them.

Find the disjunctive normal forms of the following statements:

(i) $\sim(\sim(p \leftrightarrow q) \wedge r)$

$(\wedge) \vee (\sim) \vee (\wedge)$

(ii) $p \vee (\sim p \rightarrow (q \vee (q \rightarrow \sim r)))$

(iii) $p \vee \sim(q \wedge r) \vee (p \rightarrow q)$

(iv) $(p \wedge \sim(q \vee r)) \vee (((p \wedge q) \vee \sim r) \wedge p)$

Solution

(i) $\sim(\sim(p \leftrightarrow q) \wedge r) \equiv \sim(\sim((p \rightarrow q) \wedge (q \rightarrow p)) \wedge r)$

$\equiv \sim(\sim((p \wedge q) \vee (\sim p \wedge \sim q)) \wedge r)$

$\equiv \sim((\sim p \vee \sim q) \wedge \sim(\sim p \wedge \sim q) \wedge r)$

$\equiv \sim((\sim p \vee \sim q) \wedge (p \vee q) \wedge r)$

$\equiv (\sim(\sim p \vee \sim q) \vee \sim(p \vee q) \vee \sim r)$

$\equiv (p \wedge q) \vee (\sim p \wedge \sim q) \vee \sim r$

$$(ii) p \vee (\sim p \rightarrow (q \vee (q \rightarrow \sim r)))$$

$$a \rightarrow b \equiv \sim a \vee b$$

Solution

$$p \vee (\sim p \rightarrow (q \vee (q \rightarrow \sim r))) \equiv p \vee (p \vee (q \vee (q \rightarrow \sim r)))$$

$$\equiv p \vee (p \vee (q \vee (\sim q \vee \sim r)))$$

$$\equiv p \vee (p \vee (q \vee \sim q) \vee \sim r)$$

$$\equiv T$$

Since we want it as DNF

$$\equiv p \vee p \vee q \vee \sim q \vee \sim r$$

$$\equiv p \vee q \vee \sim q \vee \sim r$$

Find the conjunction normal forms of the following statements:

$$(i) (p \wedge \sim(q \wedge r)) \vee (p \rightarrow q)$$

$$(ii) (q \vee (p \wedge q)) \wedge \sim((p \vee r) \wedge q)$$

$$(iii) (p \wedge \sim(q \wedge r)) \vee (p \rightarrow q)$$

Solution

$$(i) (p \wedge \sim(q \wedge r)) \vee (p \rightarrow q) \equiv (p \wedge (\sim q \vee \sim r)) \vee (\sim p \vee q)$$

$$\equiv (p \wedge \sim q) \vee (p \wedge \sim r) \vee (\sim p \vee q)$$

$$\equiv (p \vee p) \wedge (p \vee \sim r) \wedge (\sim q \vee p) \wedge (\sim q \vee \sim r) \vee (\sim p \vee q)$$

$$\equiv (p \vee p) \wedge (p \vee \sim r) \wedge (\sim q \vee p) \wedge (\sim q \vee \sim r \vee \sim p \vee q)$$

$$\equiv (p) \wedge (p \vee \sim r) \wedge (p \vee \sim q) \wedge (\sim q \vee q \vee \sim p \vee \sim r)$$

$$\equiv (p) \wedge (p \vee \sim r) \wedge (p \vee \sim q) \wedge (T \vee \sim p \vee \sim r)$$

$$\equiv (p) \wedge (p \vee \sim r) \wedge (p \vee \sim q)$$

$(\vee) \wedge (\vee) \wedge (\vee)$



$$\begin{aligned} \text{(ii)} \quad & (q \vee (p \wedge q)) \wedge \sim((p \vee r) \wedge q) \equiv q \wedge \sim((p \vee r) \wedge q), \text{ by absorption law} \\ & \equiv q \wedge \sim((p \vee r) \wedge q) \\ & \equiv q \wedge (\sim(p \vee r) \vee \sim q) \\ & \equiv q \wedge ((\sim p \wedge \sim r) \vee \sim q) \\ & \equiv q \wedge (\sim p \vee \sim q) \wedge (\sim r \vee \sim q) \end{aligned}$$

Obtain the principal disjunctive normal forms and the principal conjunctive normal forms of the following statements using truth tables:

(i) $(\sim p \vee \sim q) \rightarrow (p \leftrightarrow \sim q)$ (ii) $p \vee (\sim p \rightarrow (q \vee (\sim q \rightarrow r)))$ (iii) $(p \rightarrow (q \wedge r)) \wedge (\sim p \rightarrow (\sim q \wedge \sim r))$

Procedure:

- If the given statement is not a contradiction, then the disjunction of the minterms corresponding to the rows of the truth table having truth value T is the required PDNF, as it is equivalent to the given statement.
- For example if the truth value T of the statement corresponds to the truth value T, T and F for the variables p, q and r respectively, then the corresponding minterm is taken as $p \wedge q \wedge \sim r$
- If the given statement A is not a tautology, we can find the equivalent PCNF as follows:
 - We write down the PDNF of $\sim A$, which is the disjunction of the minterms corresponding to the rows of the truth table having the truth value F . Then if we find $\sim(\sim A) = A$, we will get the required PCNF of A .
 - Equivalently the PCNF is the conjunction of maxterms corresponding to the F values of A . But the max terms corresponding to T, T, F value of p, q , and r is $\sim p \vee \sim q \vee r$

$$(i) (\sim p \vee \sim q) \rightarrow (p \leftrightarrow \sim q)$$

p ↓	q	$\sim p$	$\sim q$ ↓	$\sim p \vee \sim q \equiv a$	$p \leftrightarrow \sim q \equiv b$	$a \rightarrow b$
T	T	F	F	F	F	T
T	F	F	T	T	T	T
F	T	T	F	T	T	T
F	F	T	T	T	F	F

$$\text{PDNF of } (\sim p \vee \sim q) \rightarrow (p \leftrightarrow \sim q) \equiv (p \wedge q) \vee (p \wedge \sim q) \vee (q \wedge \sim p)$$

$$\text{Now PDNF of } \sim(a \rightarrow b) \equiv (\sim p \wedge \sim q)$$

$$\text{PCNF of } (a \rightarrow b) \equiv \sim(\sim p \wedge \sim q) \equiv (p \vee q)$$

Without constructing the truth tables, find the principal disjunctive normal forms of the following statements:

$$(i)(\sim p \rightarrow q) \wedge (q \leftrightarrow p)$$

$$(ii)(p \wedge q) \vee (\sim p \wedge q) \vee (q \wedge r)$$

$$(iii) p \wedge \sim(q \wedge r) \vee (p \rightarrow q)$$

$$(iv)(q \vee (p \wedge r)) \wedge \sim((p \vee r) \wedge q)$$

Solution:

$$(i)(\sim p \rightarrow q) \wedge (q \leftrightarrow p) \equiv (p \vee q) \wedge ((q \wedge p) \vee (\sim q \wedge \sim p))$$

$$\equiv (p \vee q) \wedge ((p \wedge q) \vee \sim(p \vee q))$$

$$\equiv ((p \vee q) \wedge (p \wedge q)) \vee ((p \vee q) \wedge \sim(p \vee q))$$

$$\equiv ((p \vee q) \wedge (p \wedge q)) \vee F$$

$$\equiv (p \wedge (p \wedge q)) \vee (q \wedge (p \wedge q))$$

$$\equiv (p \wedge q) \vee (p \wedge q)$$

$$\equiv p \wedge q$$

$$\begin{aligned}
& (iii) p \wedge \sim(q \wedge r) \vee (p \rightarrow q) \equiv (p \wedge (\sim q \vee \sim r)) \vee (\sim p \vee q) \\
& \equiv (p \wedge \sim q) \vee (p \wedge \sim r) \vee (\sim p \vee q) \\
& \equiv (p \wedge \sim q) \vee (p \wedge \sim r) \vee (\sim p \wedge (\mathbf{q} \vee \sim \mathbf{q})) \vee (q \wedge (\mathbf{p} \vee \sim \mathbf{p})) \\
& \equiv (p \wedge \sim q) \vee (p \wedge \sim r) \vee (\sim p \wedge (q \vee \sim q)) \vee (q \wedge (p \vee \sim p)) \\
& \equiv (p \wedge \sim q) \vee (p \wedge \sim r) \vee (\sim \mathbf{p} \wedge \mathbf{q}) \vee (\sim p \wedge \sim q) \vee (q \wedge p) \vee (\mathbf{q} \wedge \sim \mathbf{p}) \\
& \equiv (p \wedge \sim q) \vee (p \wedge \sim r) \vee (\sim p \wedge q) \vee (\sim p \wedge \sim q) \vee (q \wedge p) \\
& \equiv ((p \wedge \sim q) \wedge (\mathbf{r} \vee \sim \mathbf{r})) \vee ((p \wedge \sim r) \wedge (\mathbf{q} \vee \sim \mathbf{q})) \vee ((\sim p \wedge q) \wedge (\mathbf{r} \vee \sim \mathbf{r})) \vee ((\sim p \wedge \sim q) \wedge (\mathbf{r} \vee \sim \mathbf{r})) \vee \\
& ((q \wedge p) \wedge (\mathbf{r} \vee \sim \mathbf{r})) \\
& \equiv (p \wedge \sim q \wedge r) \vee (p \wedge \sim q \wedge \sim r) \vee (p \wedge \sim r \wedge q) \vee (p \wedge \sim r \wedge \sim q) \vee (\sim p \wedge q \wedge r) \vee (\sim p \wedge q \wedge \sim r) \vee \\
& (\sim p \wedge \sim q \wedge r) \vee (\sim p \wedge \sim q \wedge \sim r) \vee (q \wedge p \wedge r) \vee (q \wedge p \wedge \sim r) \\
& \equiv (p \wedge \sim q \wedge r) \vee (\mathbf{p} \wedge \sim \mathbf{q} \wedge \sim \mathbf{r}) \vee (\mathbf{p} \wedge \mathbf{q} \wedge \sim \mathbf{r}) \vee (\mathbf{p} \wedge \sim \mathbf{q} \wedge \sim \mathbf{r}) \vee (\sim p \wedge q \wedge r) \vee (\sim p \wedge q \wedge \sim r) \vee \\
& (\sim p \wedge \sim q \wedge r) \vee (\sim p \wedge \sim q \wedge \sim r) \vee (p \wedge q \wedge r) \vee (\mathbf{p} \wedge \mathbf{q} \wedge \sim \mathbf{r}) \\
& \equiv (p \wedge \sim q \wedge r) \vee (p \wedge \sim q \wedge \sim r) \vee (p \wedge \sim q \wedge \sim r) \vee (\sim p \wedge q \wedge r) \vee (\sim p \wedge q \wedge \sim r) \vee (\sim p \wedge \sim q \wedge r) \vee \\
& (\sim p \wedge \sim q \wedge \sim r) \vee (p \wedge q \wedge r)
\end{aligned}$$

$$\begin{aligned}
& (iv) (q \vee (p \wedge r)) \wedge \sim((p \vee r) \wedge q) \\
& \equiv (q \vee (p \wedge r)) \wedge (\sim(p \vee r) \vee \sim q) \\
& \equiv (q \vee (p \wedge r)) \wedge ((\sim p \wedge \sim r) \vee \sim q) \\
& \equiv (q \wedge (\sim p \wedge \sim r)) \vee ((p \wedge r) \wedge (\sim p \wedge \sim r)) \vee (q \wedge \sim q) \vee ((p \wedge r) \wedge \sim q) \\
& \equiv (\sim p \wedge q \wedge \sim r) \vee (p \wedge \sim p \wedge r \wedge \sim r) \vee F \vee (p \wedge \sim q \wedge r) \\
& \equiv (\sim p \wedge q \wedge \sim r) \vee F \vee F \vee (p \wedge \sim q \wedge r) \\
& \equiv (\sim p \wedge q \wedge \sim r) \vee (p \wedge \sim q \wedge r)
\end{aligned}$$

Without constructing the truth tables, find the principal conjunctive normal forms of the following statements:

$$(i)(p \wedge q) \vee (\sim p \wedge q \wedge r)$$

$$(ii)(p \vee q) \wedge (r \vee \sim p) \wedge (q \vee \sim r)$$

$$(iii)(p \vee \sim(q \vee r)) \vee (((p \wedge q) \wedge \sim r) \wedge p)$$

$$(iv)(p \rightarrow (q \wedge r)) \wedge (\sim p \rightarrow (\sim q \wedge \sim r))$$

Solution:

$$(i)(p \wedge q) \vee (\sim p \wedge q \wedge r) \equiv ((p \wedge q) \vee \sim p) \wedge ((p \wedge q) \vee q) \wedge ((p \wedge q) \vee r)$$

$$\equiv (p \vee \sim p) \wedge (q \vee \sim p) \wedge (p \vee q) \wedge (q \vee q) \wedge (p \vee r) \wedge (q \vee r)$$

$$\equiv T \wedge (q \vee \sim p) \wedge (p \vee q) \wedge q \wedge (p \vee r) \wedge (q \vee r)$$

$$\equiv ((\sim p \vee q) \vee (r \wedge \sim r)) \wedge ((p \vee q) \vee (r \wedge \sim r)) \wedge ((q) \vee (p \wedge \sim p)) \wedge ((p \vee r) \vee (q \wedge \sim q)) \wedge ((q \vee r) \vee (p \wedge \sim p))$$

$$\equiv (\sim p \vee q \vee r) \wedge (\sim p \vee q \vee \sim r) \wedge (p \vee q \vee r) \wedge (p \vee q \vee \sim r) \wedge ((q \vee p) \vee (r \wedge \sim r)) \wedge ((q \vee \sim p) \vee (r \wedge \sim r)) \wedge (p \vee q \vee r) \wedge (p \vee \sim q \vee r) \wedge (p \vee q \vee r) \wedge (\sim p \vee q \vee r)$$

$$\equiv (\sim p \vee q \vee r) \wedge (\sim p \vee q \vee \sim r) \wedge (p \vee q \vee r) \wedge (p \vee q \vee \sim r) \wedge (p \vee q \vee r) \wedge (p \vee q \vee \sim r) \wedge (\sim p \vee q \vee r) \wedge (\sim p \vee q \vee \sim r) \wedge (p \vee q \vee r) \wedge (p \vee \sim q \vee r) \wedge (p \vee q \vee r) \wedge (\sim p \vee q \vee r)$$

$$\equiv (\sim p \vee q \vee r) \wedge (\sim p \vee q \vee \sim r) \wedge (p \vee q \vee r) \wedge (p \vee q \vee \sim r) \wedge (p \vee \sim q \vee r)$$

Theory of Inference

Introduction:

- *Inference theory* is concerned with the inferring of a *conclusion* from *certain hypothesis or basic assumptions*, called *premises*, by *applying certain principles* of reasoning called *rules of inference*.
- When a *conclusion derived* from a *set of premises* by using rules of inference, the process of such *derivation* is called a *formal proof*.
- The rules of inference are only means used to draw a conclusion from a set of premises in a *finite sequence of steps* called *arguments*.
- These rules will be given in terms of statement formulas rather than in terms of any specific statements or their truth values.
- Any *conclusion* which is *arrived* at the following these rules is called a *valid conclusion* and the *argument* is called a *valid argument*.
- The actual truth value of the premises and that of the conclusion do not play any part in the determination of the validity of the argument.

Truth table technique

- When A and B are two statement formulas, the B is said to (logically) follow A or B is a valid conclusion of the premise A , if $A \rightarrow B$ is a tautology, viz., $A \Rightarrow B$.
- Extending, a conclusion C is said to follow from a set of premises $H_1, H_2, \dots, H_n$, if $(H_1 \wedge H_2 \wedge \dots \wedge H_n) \Rightarrow C$
- If a set of premises and a conclusion are given, it is possible to determine whether the conclusion follows from the premises by constructing relevant truth tables, as explained in the following example. This method is known as truth table technique.

- Example, let us consider

- $H_1: \sim p, H_2: p \vee q, C: q$

- H_1 and H_2 are true only in the third row,

- in which case C is also true. Hence (i) is valid

- $H_1: p \rightarrow q, H_2: q, C: p$

- H_1 and H_2 are true in the first and third rows, but C is not true in the third row. Hence (ii) is not a valid conclusion

p	q	$\sim p$	$p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	T	F
F	T	T	T	T
F	F	T	F	T

Rules of Inference

- Before we give the frequently used rules of inference in the form of tautologies in a table, we state two basic rules of inference called rules P and T.
- Rule P: A premise may be introduced at any step in the derivation
- Rule T: A formula S may be introduced in the derivation, if S is tautologically implied by one or more preceding formulas in the derivation.

What are Rules of Inference for?

- Mathematical logic is often used for logical proofs. Proofs are valid arguments that determine the truth values of mathematical statements.
- An argument is a sequence of statements.
- The last statement is the conclusion and all its preceding statements are called premises (or hypothesis). The symbol “ $\therefore$ ”, (read therefore) is placed before the conclusion.
- A valid argument is one where the conclusion follows from the truth values of the premises.
- Rules of Inference provide the templates or guidelines for constructing valid arguments from the statements that we already have.

Rules of Inference

Rule in tautological form	Name of the rule
$(p \wedge q) \rightarrow p$ $(p \wedge q \Rightarrow p)$ $(p \wedge q) \rightarrow q$ $(p \wedge q \Rightarrow q)$	Simplification
$p \rightarrow (p \vee q)$ $q \rightarrow (p \wedge q)$	Addition
$((p) \wedge (q)) \rightarrow (p \wedge q)$	Conjunction
$[p \wedge (p \rightarrow q)] \rightarrow q$	Modus Ponens
$[\sim q \wedge (p \rightarrow q)] \rightarrow \sim p$	Modus tollens
$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r)$	Hypothetical Syllogism
$[(p \vee q) \wedge \sim p] \rightarrow q$	Disjunctive Syllogism
$[(p \vee q) \wedge (\sim p \vee r)] \rightarrow (q \vee r)$	Resolution
$[(p \vee r) \wedge (p \rightarrow r) \wedge (q \rightarrow r)] \rightarrow r$	Dilemma

Rules of Inference

Rule	Tautology	Name of the rule
$P \rightarrow q$ q ----- $\therefore q$	$((p \rightarrow q) \wedge p) \Rightarrow q$	Modus Ponens (Law of Detachment)
$P \rightarrow q$ $\sim q$ ----- $\therefore \sim p$	$((p \rightarrow q) \wedge \sim q) \Rightarrow (\sim p)$	Modus tollens
$P \rightarrow q$ $q \rightarrow r$ ----- $\therefore p \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r) \Rightarrow (p \rightarrow r)$	Hypothetical Syllogism (Transitivity)
$p \vee q$ $\sim p$ ----- $\therefore q$	$((p \vee q) \wedge \sim p) \Rightarrow q$	Disjunctive Syllogism

Rules of Inference

Rule	Tautology	Name of the rule
$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$	$p \Rightarrow p \vee q$	Addition
$\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$	$(p \wedge q) \Rightarrow (p)$	Simplification
$\begin{array}{l} p \\ q \\ \hline \therefore p \wedge q \end{array}$	$(p) \wedge (q) \Rightarrow p \wedge q$	Conjunction
$\begin{array}{l} p \vee q \\ \sim p \vee r \\ \hline \therefore q \vee r \end{array}$	$((p \vee q) \wedge (\sim p \vee r)) \Rightarrow (q \vee r)$	Resolution

Rules of Inference

Rule	Tautology	Name of the rule
$(p \rightarrow q) \wedge (r \rightarrow s)$ $\sim q \vee \sim s$ ----- $\therefore \sim p \vee \sim r$	$((p \rightarrow q) \wedge (r \rightarrow s)) \wedge$ $(\sim q \vee \sim s) \Rightarrow \sim p \vee \sim r$	Destructive Dilemma
$(p \rightarrow q) \wedge (r \rightarrow s)$ $p \vee r$ ----- $\therefore q \vee s$	$((p \rightarrow q) \wedge (r \rightarrow s)) \wedge$ $(p \vee r) \Rightarrow q \vee s$	Constructive Dilemma

Examples:

Addition:

Let P be the proposition, “He studies very hard” is true

Therefore – “Either he studies very hard or not he is a very bad student”. Here Q is the proposition “he is a very bad student” $P \vee Q$

Conjunction:

Let P – “He studies very hard”, Q – “He is the best boy in the class”

Therefore – “He studies very hard and he is the best boy in the class” $P \wedge Q$

Simplification:

“He studies very hard and he is the best boy in the class” $P \wedge Q$

Therefore – “He studies very hard” P

Modus Ponens:

“If you have a password, then you can log on to facebook” $P \rightarrow Q$

“You have a password”, P

Therefore – “You can log on to facebook”

Modus Tollens:

“If you have a password, then you can log on to facebook”, $P \rightarrow Q$

“You cannot log on to facebook” $\sim Q$

Therefore – “You do not have password” $\sim P$

Disjunctive Syllogism:

“The ice cream is not vanilla flavored”, $\sim P$

“The ice cream is either vanilla flavored or chocolate flavored” $P \vee Q$

Therefore – “The ice cream is chocolate flavored” Q

Hypothetical Syllogism:

“If it rains, I shall not go to school” $P \rightarrow Q$

“If I don’t go to school, I won’t need to do homework” $Q \rightarrow R$

Therefore - “If it rains, I won’t need to do homework” $P \rightarrow R$

Constructive Dilemma:

“If it rains, I will take a leave” $P \rightarrow Q$

“If it is hot outside, I will go for a shower” $R \rightarrow S$

“Either it will rain or it is hot outside” $P \vee R$

Therefore – “I will take a leave or I will go for a shower” $Q \vee S$

Destructive Dilemma:

“If it rains, I will take a leave” $P \rightarrow Q$

“If it is hot outside, I will go for a shower” $R \rightarrow S$

“Either it will not take a leave or I will not go for a shower” $\sim Q \vee \sim S$

Therefore – “Either It does not rain or it is not hot outside” $\sim P \vee \sim R$

Form of Argument

- When a set of given statements constitute a valid argument, the argument form will be presented as in the following example:
- “If it rains heavily, then traveling will be difficult”.
- “If students arrive on time, then travelling was not difficult”.
- “They arrived on time. Therefore, it did not rain heavily”

Let the statements be defined as follows:

- p : It rains heavily
- q : Travelling is difficult
- r : Students arrived on time

Now, we have to show that the premises $p \rightarrow q, r \rightarrow \sim q$ and r lead to the conclusion $\sim p$.

The form of argument given as follows show that the premises lead to the conclusion

Step No.	Statement	Reason
1.	$p \rightarrow q$	Rule P
2.	$\sim q \rightarrow \sim p$	T, contrapositive of 1
3.	$r \rightarrow \sim q$	Rule P
4.	$r \rightarrow \sim p$	T, step 2, 3 and hypothetical syllogism Rule P
5.	r	Rule P
6.	$\sim r$	T, steps 4, 5 and modus ponens

Rule CP or Rule of Conditional proof

In addition to the two basic rules of inference P and T . We have one more basic rule called *Rule CP*

If a formula s can be derived from another formula r and a set of premises, then the statement $(r \rightarrow s)$ can be derived from the set of premises alone.

The rule CP follows from the equivalence

$$(p \wedge r) \rightarrow s \equiv p \rightarrow (r \rightarrow s)$$

Note:

If the conclusion is of the form $r \rightarrow s$, we will take r as an additional premise and derive S using the given premises and r

Inconsistent Premises

A set of premises $H_1, H_2, \dots, H_n$ is said to be inconsistent, if their conjunction implies a **contradiction**

viz. If $H_1 \wedge H_2 \wedge \dots \wedge H_n \Rightarrow R \wedge \sim R$ for some formula R .

A set of premises is said to be consistent, if it is not inconsistent.

Indirect Method of Proof

The **notion of inconsistency** is used to derive a **proof** at times. This procedure is called the **indirect method of proof or proof by contradiction or reduction and absurdum**.

In order to show that a conclusion C follows from the premises $H_1, H_2, \dots, H_n$ by this method, we assume that C is false and include $\sim C$ as an additional premise

If the new set of premises is inconsistent leading to a contradiction, then the assumption that $\sim C$ is true. Thus C follows from $H_1, H_2, \dots, H_n$

For example, we prove that the premises $\sim q, p \rightarrow q$ result in the conclusion $\sim p$ by the indirect method of proof.

We now include $\sim(\sim p)$ as an additional premises. The argument form is given:

Step No.	Statement	Reason
1.	$\sim(\sim p)$	C
2.	p	T, double negation, 1
3.	$p \rightarrow q$	C
4.	$\sim q \rightarrow \sim p$	T, Contrapostive, 3
5.	$\sim q$	C
6.	$\sim p$	T, Modus ponens, 4, 5
7.	$p \wedge \sim p$	T, Conjunction, 2, 6

Thus the inclusion of $\sim C$ leads to a contradiction. Hence $\sim q, p \rightarrow q \Rightarrow \sim p$

Predicate Calculus or Predicate Logic

- In mathematics and computer programs, we encounter statements involving variable such as " $x > 10$ ", " $x = y + 5$ " and " $x + y = z$ ". These statements are neither true or false, when the values of the variables are not specified.
- The statement " x is greater than 10" has 2 parts. The first part, the **variable** x , is the subject of the **statement**. The second part "**is greater than 10**", which refers to a **property that the subject** can have, is called the **predicate**.
- We can denote the statement " x is greater than 10" by the notation $P(x)$, where P denotes the predicate "is greater than 10" and x is the variable. $P(x)$ is called the propositional function at x .
- Once a value has been assigned to the variable x , the statement $P(x)$ becomes a proposition and has a truth value.
- For example, the truth value of $P(15)\{\equiv 15 > 10\}$ and $P(5)\{\equiv 5 > 10\}$ are T and F respectively.
- The statement " $x = y + 5$ " and " $x + y = z$ " will be denoted by $P(x, y)$ and $P(x, y, z)$ respectively.
- The logic based on the analysis of predicates in any statement is called ***predicate logic or predicate calculus***

Quantifiers

- Many mathematical statements assert that a **property** is **true** for all values of a variable in a particular domain, called the ***universe of discourse***.
- Such a statement is expressed using a **universal quantification**. The universal quantification of $P(x)$ is the statement.
- " $P(x)$ is true for all values of x in the universe of discourse" and is denoted by the notation $(x)P(x)$ or $\forall xP(x)$.
- The proposition $(x)P(x)$ or $\forall xP(x)$ is read as "for all x , $P(x)$ " or "for every x , $P(x)$ ".
- The symbol $\forall$ is called the universal quantifier.

Existential Quantifier

- The existential quantification of $P(x)$ is the proposition
- “There exists at least one x (or an x) such that $P(x)$ is true” and is denoted by the notation $\exists xP(x)$.
- The symbol $\exists$ is called **existential quantifier**.
- The proposition $\exists xP(x)$ is read as “For some x , $P(x)$ ”

Negation of a Quantified Expression

- If $P(x)$ is the statement " x has studied computer programming", then $\forall x P(x)$ means that "every student (in the class) has studied computer programming".
- The negation of this statement is "It is not the case that every student in the class has studied computer programming" or equivalently "There is a student in the class who has not studied computer programming" which is denoted by $\exists x \sim P(x)$.
- Thus, we see that $\sim \forall x P(x) \equiv \exists x \sim (P(x))$
- Similarly, $\exists x P(x)$ means that "There is a student in the class who has studied computer programming". The negation of this statement is "Every student in this class has not studied computer programming", which is denoted by $\forall x \sim P(x)$. Thus we get
- $\sim (\exists x P(x)) \equiv \forall x (\sim P(x))$

- Further we note that $\sim \forall x P(x)$ is true, when there is an x for which $P(x)$ is false and false when $P(x)$ is true for every x , since

$$\sim \forall x P(x) \equiv \exists x \sim P(x) \equiv \sim P(x_1) \vee \sim P(x_2) \vee \cdots \vee \sim P(x_n)$$

- $\sim \exists x P(x)$ is true, when $P(x)$ is false for every x and false when there is an x for which $P(x)$ is true,

$$\sim \exists x P(x) \equiv \forall x \sim P(x) \equiv \sim P(x_1) \wedge \sim P(x_2) \wedge \cdots \wedge \sim P(x_n)$$

Nested (More than one) Quantifiers

There are situation when quantifiers occur in combinations in respect of 1-place or n -place predicate formulas (i.e., propositional functions containing 1 or n variables). For example let us consider a 2-place predicate formula $P(x, y)$

$$\text{Now, } \forall x \forall y P(x, y) \equiv \forall x [\forall y P(x, y)] \equiv \forall y [\forall x P(x, y)] \quad \text{----- (1)}$$

$$\text{And } \exists x \exists y P(x, y) \equiv \exists x [\exists y P(x, y)] \equiv \exists y [\exists x P(x, y)] \quad \text{----- (2)}$$

From the meaning of quantifiers and by (1) and (2), the following simplifications hold good:

$$\forall x \forall y P(x, y) \Rightarrow (\exists y) \forall x [P(x, y)] \Rightarrow \forall x \exists y P(x, y)$$

$$\forall y \forall x P(x, y) \Rightarrow (\exists x) \forall y [P(x, y)] \Rightarrow \forall y \exists x P(x, y)$$

Free and Bound Variables

- When a **quantifier** is **used on a variable** x or when we have to assign a value to this variable to get a proposition. The occurrence of the variable is said to be bound or the variable is said to be a **bound variable**.
- An **occurrence of a variable** that is **not bound** by a **quantifier** or **that is set equal to** a particular value is said to be **free**.
- The part of the logical expression or predicate formula to which a quantifier is applied is called the **scope of the quantifier**.

Examples

Predicate formula	Bound Variable and Scope	Free Variable
$\forall x P(x, y)$	$x: P(x, y)$	y
$\forall x (P(x) \rightarrow Q(x))$	$x: P(x) \rightarrow Q(x)$	-
$\forall x (P(x) \rightarrow E(y)Q(x, y))$	$x: P(x) \rightarrow E(y)Q(x, y)$ $y: Q(x, y)$	-
$\forall x (P(x) \wedge Q(x)) \vee \forall y R(y)$	$x: P(x) \wedge Q(x)$ $y: R(y)$	-
$\exists x P(x) \wedge Q(x)$	First $x: P(x)$	Second x

Valid Formulas and Equivalences

- Let A and B be any two predicate formulas defined over a common universe of discourse E .
- When each of the variables appearing in A and B is replaced by any element (object name) of the universe E , if the resulting statements have the same truth values, then A and B are said to be **equivalent** to each other over E and denoted as $A \equiv B$ or $A \Leftrightarrow B$ over E .
- If E is arbitrary, we simply say that A and B are equivalent and denote it as $A \equiv B$ or $A \Leftrightarrow B$.
- Generally, logically valid formulas in predicate calculus can be obtained from tautologies of propositional calculus by replacing primary propositions such as p, q, r by propositional functions.
- For example, $p \vee \sim p \equiv T$ and $(p \rightarrow q) \leftrightarrow (\sim p \vee q) \equiv T$ are tautologies in statement calculus.
- If we replace p by $\forall x R(x)$ and q by $\exists x S(x)$ in the above, we get the following valid formulas in predicate calculus
- $(\forall x R(x)) \vee (\sim \forall x R(x)) \equiv T$
- $((\forall x R(x)) \rightarrow (\exists x S(x))) \leftrightarrow ((\sim \forall x R(x)) \vee \exists x S(x)) \equiv T$

More generally, all the implications and equivalences of the statement calculus can also be considered as implications and equivalences of the predicate calculus if we replace elementary statements by primary predicate formulas. For example,

From $\sim(\sim p) \Leftrightarrow p$, we get $\sim(\sim P(x)) \equiv P(x)$ -----(1)

From $p \wedge q \equiv q \wedge p$, we get $P(x) \wedge Q(x, y) \equiv Q(x, y) \wedge P(x)$ -----(2)

From $p \rightarrow q \equiv \sim p \vee q$, we get $P(x) \rightarrow Q(x) \equiv \sim P(x) \vee Q(x)$ -----(3)

(1), (2) and (3) are some examples for valid formulas in predicate calculus.

Apart from the types of valid formulas given above, there are other valid formulas also which involve quantifiers.

Such valid formulas are obtained by using the inference theory of predicate logic.

Inference Theory or Predicate Calculus

Derivations of formal proof in predicate calculus are done mostly in the same way as in statement calculus, using implications and equivalences, provided that the statement formulas are replaced by predicate formulas.

Also the three basic rules P, T and CP of inference theory used in statement calculus can also be used in predicate calculus. Moreover, the indirect method of proof can also be used in predicate calculus

Apart from the above rules of inference, we require certain additional rules to deal with predicate formulas involving quantifiers. If it becomes necessary to eliminate quantifiers during the course of derivation. We require two rules of specification, called *US* and *ES* rules.

Once the quantifiers are eliminated, the derivation is similar to that in statement calculus. If it becomes necessary to quantify the desired conclusion.

We require two rules of generalisation, called UG and EG rules.

Rule US: Universal Specification is the rule of inference which states that one can conclude that $P(c)$ is true, if $\forall x P(x)$ is true, where c is an arbitrary member of the universe of discourse. This rule is also called the universal instantiation

Rule ES: Existential Specification is the rule which allows us to conclude that $P(c)$ is true, if $\exists x P(x)$ is true, where c is not an arbitrary member of the universe, but one for which $P(c)$ is true. Usually we will not know what c is but know that it exists. Since it exists, we may call it c . This rule is also called the existential instantiation.

Rule UG: Universal Generalisation is the rule which states that $\forall x P(x)$ is true, if $P(c)$ is true, where c is an arbitrary member (not a specific member) or the universe of discourse

Rule EG: Existential Generalisation is the rule that is used to conclude that $\exists x P(x)$ is true when $P(c)$ is true, where c is a particular member of the universe of discourse.

Example: Let us consider the following “Famous Socrates argument” which is given by:

All men are mortal

Socrates is a man.

Therefore Socrates is a mortal

Solution:

$H(x)$: x is a man

$M(x)$: x is a mortal

s : Socrates

With these symbolic notations, the problem becomes

$$\forall x \left((H(x)) \rightarrow M(x) \right) \wedge H(s) \Rightarrow M(s)$$

The derivation of the proof is as follows:

Step No.	Statement	Reason
1	$\forall x (H(x) \rightarrow M(x))$	P
2	$H(S) \rightarrow M(S)$	$US, 2$
3	$H(S)$	P
4	$M(S)$	T, 2, 3, Modus Ponen

Problem:

1. Find whether the conclusion C follows from the premises H_1, H_2, H_3 in the following cases, using truth table technique:

(i) $H_1: \sim p, H_2: p \vee p, C: p \wedge q$

(ii) $H_1: p \vee q, H_2: p \rightarrow r, H_3: q \rightarrow r, C: r$

Solution

(i)

p	q	$H_1 \equiv \sim p$	$H_2 \equiv p \vee q$	$H_1 \wedge H_2$	$C \equiv p \wedge q$
T	T	F	T	F	T
T	F	F	T	F	F
F	T	T	T	T	F
F	F	T	F	F	F

H_1 and H_2 and hence $H_1 \wedge H_2$ are true in the third row, in which C is false.

Hence C does not follow from H_1 and H_2

(ii) $H_1: p \vee q, H_2: p \rightarrow r, H_3: q \rightarrow r, C: r$

p	q	r	$H_1 \equiv p \vee q$	$H_2 \equiv p \rightarrow r$	$H_3 \equiv q \rightarrow r$	$H_1 \wedge H_2 \wedge H_3$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	T	T	T	T
T	F	F	T	F	T	F
F	T	T	T	T	T	T
F	T	F	T	T	F	F
F	F	T	F	T	T	F
F	F	F	F	T	T	F

$H_1 \wedge H_2 \wedge H_3$ are true in first, third and fifth rows in which r is also true

Hence C follows from H_1, H_2 and H_3

2. Show that $(t \wedge s)$ can be derived from the premises $p \rightarrow q, q \rightarrow \sim r, r, p \vee (t \wedge s)$.

Solution:

Step No.	Statement	Reason
1.	$p \rightarrow q$	Rule P
2.	$q \rightarrow \sim r$	Rule P
3.	$p \rightarrow \sim r$	T, 1, 2 and Hypothetical Syllogism
4.	$r \rightarrow \sim p$	T, 3 and Contrapositive ($p \rightarrow q \equiv \sim q \rightarrow \sim p$)
5.	r	Rule P
6.	$\sim p$	T, 4, 5 and Modus ponens
7.	$p \vee (t \wedge s)$	Rule P
8.	$t \wedge s$	T, 6, 7 and Disjunctive Syllogism

3. Show that $(a \vee b)$ follows logically from the premises $(p \vee q)$, $(p \vee q) \rightarrow \sim r$, $\sim r \rightarrow (s \wedge \sim t)$ and $(s \wedge \sim t) \rightarrow (a \vee b)$.

Solution

Step No.	Statement	Reason
1.	$(p \vee q) \rightarrow \sim r$	Rule P
2.	$\sim r \rightarrow (s \wedge \sim t)$	Rule P
3.	$(p \vee q) \rightarrow (s \wedge \sim t)$	T, 1, 2 and Hypothetical Syllogism
4.	$p \vee q$	Rule P
5.	$s \wedge \sim t$	T, 3, 4 and Modus Ponens
6.	$(s \wedge \sim t) \rightarrow (a \vee b)$	Rule P
7.	$a \vee b$	T, 5, 6 and Modus Ponens

4. Show that $(p \rightarrow q) \wedge (r \rightarrow s), (q \rightarrow t) \wedge (s \rightarrow u), \sim(t \wedge u)$ and $(p \rightarrow r) \Rightarrow \sim p$.

Solution:

Step No.	Statement	Reason
1.	$(p \rightarrow q) \wedge (r \rightarrow s)$	Rule P
2.	$p \rightarrow q$	T, 1 and simplification
3.	$r \rightarrow s$	T, 1 and simplification
4.	$(q \rightarrow t) \wedge (s \rightarrow u)$	Rule P
5.	$q \rightarrow t$	T, 4 and simplification
6.	$s \rightarrow u$	T, 4 and simplification
7.	$p \rightarrow t$	T, 2, 5 and hypothetical syllogism
8.	$r \rightarrow u$	T, 3, 6 and hypothetical syllogism
9.	$p \rightarrow r$	Rule P
10.	$p \rightarrow u$	T, 9, 8 and Hypothetical syllogism
11.	$\sim t \rightarrow \sim p$	T, 7 and contrapositive
12.	$\sim u \rightarrow \sim p$	T, 10 and contrapositive
13.	$(\sim t \vee \sim u) \rightarrow \sim p$	T, 11, 12 and $(a \rightarrow b), (c \rightarrow b) \Rightarrow (a \vee c) \rightarrow b$
14.	$\sim(t \wedge u) \rightarrow \sim p$	T, 13 and De Morgan's Law
15.	$\sim(t \wedge u)$	Rule P
16.	$\sim p$	T, 14, 15 and Modus Ponens

Homework

5. Show that $(a \rightarrow b) \wedge (a \rightarrow c), \sim(b \wedge c), (d \vee a) \Rightarrow d$

6. Give a direct proof for the implication $p \rightarrow (q \rightarrow s), (\sim r \vee p), q \Rightarrow (r \rightarrow s)$

Solution

Step No.	Statement	Reason
1.	$(\sim r \vee p)$	Rule P
2.	$r \rightarrow p$	T, 1 and equivalence of (1)
3.	$p \rightarrow (q \rightarrow s)$	Rule P
4.	$r \rightarrow (q \rightarrow s)$	T, 2, 3 and hypothetical syllogism
5.	$\sim r \vee (\sim q \vee s)$	T, 4 and equivalence of (4)
6.	q	Rule P
7.	$q \wedge (\sim r \vee \sim q \vee s)$	T, 5, 6 and Conjunction
8.	$q \wedge (\sim r \vee s)$	T, 7, 8 and negation and domination laws
9.	$\sim r \vee s$	T, 8 and simplification
10.	$r \rightarrow s$	T, 9 and equivalence of (9)

7. Derive $p \rightarrow (q \rightarrow s)$ using the CP-rule (if necessary) from the premises $p \rightarrow (q \rightarrow r)$ and $q \rightarrow (r \rightarrow s)$

Solution:

We shall assume p as an additional premise. Using p and the two given premises, we will derive $(q \rightarrow s)$. Then, by CP-rule, $p \rightarrow (q \rightarrow s)$ is deemed to have been derived from the two given premises.

Step No.	Statement	Reason
1.	p	P (additional)
2.	$p \rightarrow (q \rightarrow r)$	P
3.	$q \rightarrow r$	T, 1, 2 and Modus Ponens
4.	$\sim q \vee r$	T, 3 and equivalence of (3)
5.	$q \rightarrow (r \rightarrow s)$	P
6.	$\sim q \vee (r \rightarrow s)$	T, 5 and equivalence of (5)
7.	$\sim q \vee (r \wedge (r \rightarrow s))$	T, 4, 6 and distributive law
8.	$\sim q \vee s$	T, 7 and modus ponens
9.	$q \rightarrow s$	T, 8 and equivalence of (8)
10.	$p \rightarrow (q \rightarrow s)$	T, 9 and CP-rule

8. Use the indirect method to show that $r \rightarrow \sim q, r \vee s, s \rightarrow \sim q, p \rightarrow q \Rightarrow \sim p$

Solution:

To use indirect method, we will include $\sim(\sim p) \equiv p$ as an additional premise and prove a contradiction.

Step No.	Statement	Reason
1.	p	P
2.	$p \rightarrow q$	P
3.	q	T, 1, 2 and Modus Ponens
4.	$r \rightarrow \sim q$	P
5.	$s \rightarrow \sim q$	P
6.	$(r \vee s) \rightarrow \sim q$	T, 4, 5 and equivalence
7.	$r \vee s$	P
8.	$\sim q$	T, 6, 7 and Modus Ponens
9.	$q \wedge \sim q$	T, 3, 8 and Conjunction
10.	F	T, 9 and negation law

9. Show that b can be derived from the premises $a \rightarrow b, c \rightarrow b, d \rightarrow (a \vee c), d$ by the indirect method.

Solution:

Let us include $\sim b$ as an additional premise and prove a contradiction

Step No.	Statement	Reason
1.	$a \rightarrow b$	P
2.	$c \rightarrow b$	P
3.	$(a \vee c) \rightarrow b$	T, 1, 2 and Equivalence
4.	$d \rightarrow (a \vee c)$	P
5.	$d \rightarrow b$	T, 3, 4 and hypothetical syllogism
6.	d	P
7.	b	T, 5, 6 and modus ponens
8.	$\sim b$	P (additional)
9.	$b \wedge \sim b$	T, 7, 8 and conjunction
10.	F	T, 9 and negation law

10. Using indirect method of proof, derive $p \rightarrow \sim s$ from the premises $p \rightarrow (q \vee r)$, $q \rightarrow \sim p$, $s \rightarrow \sim r$, p .

Solution: Let us include $\sim(p \rightarrow \sim s)$ as an additional premise and prove a contradiction.

$$\sim(p \rightarrow \sim s) \equiv \sim(\sim p \vee \sim s) \equiv p \wedge s$$

Step No.	Statement	Reason
1.	$p \rightarrow (q \vee r)$	P (additional)
2.	p	P
3.	$(q \vee r)$	T, 1, 2 and Modus ponens
4.	$p \wedge s$	P (additional)
5.	s	T, 4 and simplification
6.	$s \rightarrow \sim r$	P
7.	$\sim r$	T, 5, 6 and modus ponens
8.	q	T, 3, 7 and disjunctive syllogism
9.	$q \rightarrow \sim p$	P
10.	$\sim p$	T, 8, 9 and modus ponens
11.	$p \wedge \sim p$	T, 2, 10 and conjunction
12.	F	T, 11 and negation law

11. Prove the premises $p \rightarrow q, q \rightarrow r, s \rightarrow \sim r$ and $q \wedge s$ are inconsistent.

Solution:

If we derive a contradiction by using the given premises, it means that they are inconsistent.

Step No.	Statement	Reason
1.	$p \rightarrow q$	P
2.	$q \rightarrow r$	P
3.	$p \rightarrow r$	T, 1, 2 and hypothetical syllogism
4.	$s \rightarrow \sim r$	P
5.	$r \rightarrow \sim s$	T, 4 and contradiction
6.	$q \rightarrow \sim s$	T, 2, 5 and hypothetical syllogism
7.	$\sim q \vee \sim s$	T, 6 and equivalence of (6)
8.	$\sim(q \wedge s)$	T, 7 and De Morgan's law
9.	$q \wedge s$	P
10.	$(q \wedge s) \wedge \sim(q \wedge s)$	T, 8, 9 and conjunction
11.	F	T, 10 and negation law

Hence the given premises are inconsistent

12. Prove the premises $a \rightarrow (b \rightarrow c)$, $d \rightarrow (b \wedge \sim c)$ and $(a \wedge d)$ are inconsistent.

Solution

Step No.	Statement	Reason
1.	$a \wedge d$	P
2.	a	T, 1 and simplification
3.	d	T, 1 and simplification
4.	$a \rightarrow (b \rightarrow c)$	P
5.	$b \rightarrow c$	T, 2, 4, and modus ponens
6.	$\sim b \vee c$	T, 5 and equivalence of (5)
7.	$d \rightarrow (b \wedge \sim c)$	P
8.	$\sim(b \wedge \sim c) \rightarrow \sim d$	T, 7 and contrapositive
9.	$\sim b \wedge c \rightarrow \sim d$	T, 8 and equivalence
10.	$\sim d$	T, 6, 9 and modus ponens
11.	$d \wedge \sim d$	T, 1, 10 and conjunction
12.	F	T, 11 and negation law

Hence the given premises are inconsistent

13. Construct an argument to show that the following premises imply the conclusion “it rained”.

“If it does not rain or if there is no traffic dislocation, then the sports day will be held and the cultural programme will go on”: “If the sports day is held, the trophy will be awarded” and “The trophy was not awarded”

Solution:

Let us symbolize the statement as follows:

p : It rains

q : There is traffic dislocation

r : sports day will be held

s : Cultural programme will go on

t : the trophy will be awarded

Then we have to prove that $\sim p \vee \sim q \rightarrow r \wedge s, r \rightarrow t, \sim t \Rightarrow p$

Step No.	Statement	Reason
1.	$\sim p \vee \sim q \rightarrow r \wedge s$	P
2.	$(\sim p \rightarrow (r \wedge s)) \wedge (\sim q \rightarrow (r \wedge s))$	T, 1 and equivalence $(a \vee b) \rightarrow c \equiv (a \rightarrow c) \wedge (b \rightarrow c)$
3.	$(\sim p \rightarrow (r \wedge s))$	T, 1 and simplification
4.	$\sim(r \wedge s) \rightarrow p$	T, 3 and contrapositive of (3)
5.	$r \rightarrow t$	P
6.	$\sim t \rightarrow \sim r$	T, 5 and contrapositive of (5)
7.	$\sim t$	P
8.	$\sim r$	T, 6, 7 and modus ponens
9.	$\sim r \vee \sim s$	T, 8 and addition
10.	$\sim(r \wedge s)$	T, 9 and De Morgan's law
11.	p	T, 4, 10 and modus ponens

14. Show that the following set of premises is inconsistent:

If Ram gets his degree, he will go for a job.

If he goes for a job, he will get married soon

If he goes for higher study, he will not get married

Rama gets his degree and goes for higher study.

Solution:

Let the statement be symbolized as follows:

p : Rama gets his degree

q : He will go for a job

r : He will get married soon

s : He goes for higher study

Then prove that $p \rightarrow q, q \rightarrow r, s \rightarrow \sim r, p \wedge s$ are inconsistent

15. Express each of the following statements using mathematical and logical operations, predicates and quantifiers, where the universe of discourse consists of all computer science students/ mathematics courses.

(a) Every computer science students need a course in mathematics

(b) There is a student in this class who owns a personal computer

(c) Every student in this class has taken at least one mathematics course

(d) There is a student in this class who has taken at least one mathematics course

Solution:

(a) Let $M(x) \equiv 'x \text{ needs a course in mathematics}'$, where the universe of discourse consists of all computer science students. Then $\forall x M(x)$

(b) Let $P(x) \equiv 'x \text{ owns a personal computer}'$, where the universe consists of all students in this class, then $\exists x P(x)$

(c) Let $Q(x, y) \equiv 'x \text{ has taken } y'$, where the universe of x consists of all students in this class and that of y consists of all mathematics courses. Then $\forall x \exists y Q(x, y)$

(d) Using the same assumptions as in (c), we have $\exists x \exists y Q(x, y)$

16. Show that the premises “one student in the class knows how to write programs in JAVA” and “Everyone who knows how to write programs in JAVA can get a high-paying job” imply the conclusion “someone in this class can get a high-paying job”.

Solution:

Let $C(x)$ represent “x is in this class”

$J(x)$ represent “x knows JAVA Programming”

$H(x)$ represent “x can get a high paying job”

Then the given premises are $\exists x (C(x) \wedge J(x))$ and $\forall x (J(x) \rightarrow (H(x)))$. The conclusion is $\exists x (C(x) \wedge H(x))$

Step No.	Statement	Reason
1.	$\exists x (C(x) \wedge J(x))$	P
2.	$C(a) \wedge J(a)$	ES and 1
3.	$C(a)$	T, 2 and simplification
4.	$J(a)$	T, 2 and simplification
5.	$\forall x (J(x) \rightarrow (H(x)))$	P
6.	$J(a) \rightarrow H(a)$	US and 5
7.	$H(a)$	T, 4 and 6, Modus ponens
8.	$C(a) \wedge H(a)$	T, 3, 7 and Conjunction
9.	$\exists x (C(x) \wedge H(x))$	EG and 8

17. Express the negations of the following statements using quantifiers and English:

(a) If the teacher is absent, then some students do not keep quiet.

(b) All the students keep quiet and the teacher is present

(c) Some of the students do not keep quiet or the teacher is absent

(d) No one has done every problem in the exercise.

Solution:

(a) Let T represent the presence of the teacher and $Q(x)$ represent “ x ” keeps quiet

Then the given statement $\sim T \rightarrow \exists x \sim Q(x) \equiv \sim T \rightarrow \sim \forall x Q(x) \equiv T \vee \sim \forall x Q(x)$

Negation: $\sim(T \vee \sim \forall x Q(x)) \equiv \sim T \wedge \forall x Q(x)$

i.e The teacher is absent and all the students keep quiet.

(b) All the students keep quiet and the teacher is present

$$\forall x Q(x) \wedge T$$

The negation is $\sim[\forall x Q(x) \wedge T] \equiv \exists x \sim Q(x) \vee \sim T$

i.e. some of the students do not keep quiet or teacher is absent

(c) Some of the students do not keep quiet or teacher is absent: $\exists x \sim Q(x) \vee \sim T$

Negation : $\forall x Q(x) \wedge T \Rightarrow$ All the students keep quiet and the teacher is present

(d) No one has done every problem in the exercise.

Let $D(x, y)$ represent x has done problem y

The given statement is $(\sim \exists x) (\forall y) D(x, y)$

Negation: $\sim((\sim \exists x) (\forall y) D(x, y)) \equiv (\exists x)(\forall y) D(x, y)$

Some one has done every problem in the exercise

Tutorial Problems

1. Prove that $H_1: \sim q, H_2: p \rightarrow q, C: \sim p$
2. Prove that $p \vee q, p \rightarrow r, q \rightarrow s \Rightarrow s \vee r$ using direct method
3. Prove that $p \rightarrow q, q \rightarrow r, \sim(p \wedge r), p \vee r \Rightarrow r$ using indirect method
4. Prove the following by using the CP rule $(p \vee q) \rightarrow r \Rightarrow (p \wedge q) \rightarrow r$
5. Prove that $\forall x (P(x) \rightarrow (Q(y) \wedge R(x))), \exists x P(x) \Rightarrow Q(y) \wedge \exists x (P(x) \wedge R(x))$
6. Prove the implication $\forall x (P(x) \rightarrow Q(x)), \forall x (R(x) \rightarrow \sim Q(x)) \Rightarrow \forall x (R(x) \rightarrow \sim P(x))$
7. Show that $\forall x (P(x) \vee Q(x)) \Rightarrow \forall x P(x) \vee \exists x Q(x)$ using the indirect method.
8. If $A(x)$: x is animal, $B(x)$: x is black and $C(x)$: x is a cat, translate the following in words:
(a) $\forall x [c(x) \rightarrow A(x)]$ (b) $\exists x [c(x) \wedge B(x)]$